# SP3172 - Report Draft 1

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**Abstract**

Extreme weather, potentially intensified by climate change, makes Singapore vulnerable to both inland (caused by excessive rainfall) and coastal floods (caused by increase in sea level). The combined effect of high speed winds over the South China Sea (SCS) and spring tides have been linked to both coastal and flash floods. These SCS winds are characteristic of the boreal winter monsoon cold surge (CS) events that occur over the Maritime Continent during the northeast (NE) monsoon season. The primary motivation of this study is to quantify the role that CS events play on extreme tide events. The study uses research-quality water level from tide gauges, obtained from the University of Hawaii Sea Level Centre from 1990 – 2013 along the Western Maritime Continent. Further analysis of the data has allowed us to quantify 'extreme tides' (water levels above the 99$^{\text{th}}$ percentile), the co-occurrences of CS events and 'extreme tides', along with the average increase in water levels associated with CS events. Henceforth, we develop and test a model based on water level, wind speed and pressure level anomalies, along with other climatic variables associated with CS events to predict the daily maximum and extreme water levels. The development of this model aims to improve our understanding of surge-induced flash floods, with implications for disaster preparedness in South-East Asia, thus aiding government agencies in risk mitigation for vulnerable areas and early warning systems.

# Introduction

Singapore, located at the southern tip of peninsular Malaysia, has a tropical climate, which can be divided into four primary climatic periods **tkalich˙storm˙2013**(cite V3 report). Among them, the northeast (NE) monsoon between November-February and southwest (SW) monsoon provide the area with rainfall. Between November-February, northern continental Asia faces extremely low temperatures, thus causing cold air surges towards the South China Sea. The cold air is further warmed by the seas, thus bringing rainfall to the tropics, popularly known as monsoon surges (V3 report). These cold air surges are a key feature of the boreal winter monsoons, further strengthened by the presence of the Madden Julian Oscillation (Chang et al., 2005, Lim et al., 2017).

Coastal low-lying areas in peninsular Malaysia are susceptible to flooding caused by a coincidence of high tides and extreme precipitation (Lian et al., 2017), which have often overwhelmed urban and rural drainage systems, as was observed in three separate surge events between January to March in 2025. Local mean sea levels (MSL), measured using tide gauge stations in coastal regions, are not only affected by large=scale global and regional changes (such as mass changes in ice sheets and glaciers, ocean currents, density), but also by tides and surges (V3, 2022). While tides are deterministic and can be predicted by the gravitational attraction based on the positions of the sun and the moon (Hicks, 2006), local and regional climatic effects tend to be more stochastic, and cannot be easily predicted.

# Data and Methodology

- Tides have two main signals - deterministic and stochastic.
- The deterministic signals can be predicted by the phase of the celestial bodies around the Earth at that position of the tide gauge
- Removing this signal from our net tide data will allow us to obtain the pure stochastic signals - i.e water levels influenced by winds, rainfall, variation in temperature, etc.
- The stochastic signal is of concern to us.
- We tried using two forms of data - one of hourly frequency from the University of Hawaii Sea Level Centre (citation) and the other was of six minute frequency from Maritime Port Authority.
- Both data sets were processed through the UTide Package (citation).
- Reason for choosing both - MPA gives us data around singapore, while UHSLC gives us data around Peninsular Malaysia. Taking data from gives us some level of reproducibiliity around the area.
- Used the QA/QC manual to clean the data (cite the manual)
- After the QA/QC, the cleaned data was passed to the software through which we can separate the tidal and stochastic signals.
- Taking ERA5 re-analysis wind-speed and mean sea level pressure data, we calculate their daily area-weighted average, as has been used in Lim et al (2017) (look up citation format), and in Singapore's third climate change study V3 (citation needed) using CDO (citation needed).
- Using these processed data sets we were able to develop a multivariate linear regression model (will test other models and then check for robustness - I think this analysis should be in the next section) which allows us to predict the correlation between extreme water levels that result in nuisance flooding events and the associated weather phenomena.

# Results and Discussion

This is for the Ridge regression model - to handle datasets, whose dependent variables have a high collinearity.

| Location | RMSE | R-squared |
|---|---|---|
| Cendering | 118.4696 | 0.2578 |
| Geting | 159.6380 | 0.1236 |
| Kuantan | 102.3150 | 0.2087 |
| Johor Baharu | 104.1754 | 0.2697 |
| Sedili | 124.1224 | 0.1677 |
| Tioman | 96.9017 | 0.2386 |
| Tanjong Pagar | 104.4715 | 0.1823 |

This is for a quadratic fit model.

| Location | RMSE | R-squared |
|---|---|---|
| Cendering | 110.267 | 0.290 |
| Geting | 156.548 | 0.195 |
| Kuantan | 101.374 | 0.281 |
| Johor Baharu | 96.557 | 0.289 |
| Sedili | 114.041 | 0.280 |
| Tioman | 99.418 | 0.279 |
| Tanjong Pagar | 94.459 | 0.224 |

## Conclusion

## Future Work

# References


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